X01 MATERIAL RESTRICTIONS | P

Intent : To reduce or eliminate human exposure to building materials known to be hazardous.

Summary : This WELL feature restricts widely known hazardous ingredients in newly installed building materials, specifically asbestos, mercury and lead.

Issue : Historical use of hazardous materials in construction, specifically asbestos, mercury and lead, have presented serious and negative health impacts on humans. Disease caused by exposure to these chemicals, including asbestosis, developmental issues in children[1] and various forms of cancer still affect millions of people.[2] While these compounds have been restricted or banned in buildings in many countries, they still pose as threats in countries that have not enacted the necessary limitations. As a result, exposure to these hazardous materials should be limited and, if possible, eliminated.

Solutions : Asbestos has been fully or partially banned for buildings in most countries and alternatives are widely available. Lead content in materials that may expose humans to its aspiration and ingestion is also restricted in many national regulations.[3] Use of products where lead content is minimized or not added can significantly reduce leaching from pipes into drinking water. Eliminating the use of compact fluorescent light bulbs (CFLs) removes a potential pathway for exposure to mercury.

PART 1 RESTRICT ASBESTOS

For All Spaces:

For newly installed or applied products within the project boundary, the following requirement is met:

a. The following product categories do not contain over 1,000 ppm of asbestos by weight or area:
   1. Thermal protection, including all insulation (lagging) applied to pipes, fittings, boilers, tanks and ducts.
   2. Acoustic treatments.
   3. Sheathing.
   4. Roofing and siding.
   5. Fire and smoke protection.
   6. Joint protection.
   7. Plaster and gypsum board.
   8. Ceilings.
   9. Resilient flooring.

PART 2 RESTRICT MERCURY

For All Spaces:

The following requirements are met:

a. Newly installed fluorescent, metal halide and sodium lamps, if present, meet one of the following:
   1. RoHS restrictions.[4]
   2. The following specifications:[5]

Fluorescent Lamp	Maximum Mercury Content
Compact, integral ballast	3.5 mg
Compact, no-integral ballast	3.5 mg
T-5, circular	9 mg
T-5, linear	2.5 mg
T-8, eight-foot	10 mg
T-8, four-foot	3.5 mg
T-8, U-bent	6 mg
High-Pressure Sodium Lamp	**Maximum Mercury Content**
400 W or less	10 mg
Over 400 W	32 mg

b. Newly installed fire alarm notification and initiating devices (e.g. strobes, pull stations); environmental, HVAC, occupancy and motion sensors and meters; and relays, thermostats and load break switches meet one of the following:
   1. RoHS restrictions.[4]
   2. Products contain no more than 0.1% (1000 ppm) of mercury by weight.

PART 3 RESTRICT LEAD

For All Spaces:

1: Paints and electronics

The following requirements are met:

a. Newly installed fire alarms notification and initiating devices (e.g. strobes, pull stations); environmental, HVAC, occupancy and motion sensors and meters; and relays, thermostats and load break switches meet one of the following:
   1. RoHS restrictions.[4]
   2. Products contain no more than 0.1% (1000 ppm) of lead by weight.

b. Newly installed paints applied as finishes within the project boundary meet at least one of the following criteria:
   1. Paints have a lead concentration of 100 ppm (0.01%) by weight or below.
   2. Paints have no added lead carbonates and lead sulfates.
   3. Paints are deemed free of lead or with no added lead by an ISO 14024-compliant (Type 1) Ecolabel, or a voluntary third-party certification program recognized by the local government where the building is located.
   4. Paints meet Feature X08: Materials Optimization Part 1, Option 1 'Materials Selection' OR Part 2.

2: Drinking water pipes, fittings and solder

Pipes, fixtures, fittings and solder newly installed or applied within the project boundary intended for drinking water distribution and delivery meet at least one of the following:

a. The product is approved for use with drinking water by a local government authority or by a government-authorized certification body.
b. The product has a weighted wetted average of 0.25% of lead or less, verified by a third party, or is labeled as ANSI/NSF 372-compliant.

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X02 INTERIOR HAZARDOUS MATERIALS MANAGEMENT | P

Intent : Manage risks of human exposure to hazardous materials ubiquitously used in past construction practices.

Summary : This WELL feature requires the application of practices to manage exposure risks of the hazardous building materials asbestos, lead and polychlorinated biphenyls (PCBs).

Issue : Some materials that were commonly used in construction due to their mechanical, chemical or visual properties has shown to be toxic, and are now banned or confined to limited production across the world. However, their ubiquitous presence still haunts our built and natural environments and can cause disease upon exposure. Asbestos, lead and polychlorinated biphenyls (PCBs)-containing materials are emblematic examples. Asbestos, a naturally present, chemically resistant material found in old adhesives, insulation and sheeting is a known toxicant and carcinogen. Exposure to asbestos-containing dusts is the main cause of mesothelioma and responsible of over 200,000 deaths annually.[1] Lead is found in paints for increasing their durability and impermeability. If ingested, absorbed or breathed-in, it accumulates in blood, tissues and bones, potentially disrupting body functions and impairing the intellectual development of children and the unborn during pregnancy.[2] PCBs were used in caulk and electrical equipment, but due to their persistence in the environment, bioaccumulation in foods and carcinogenicity, their production is banned worldwide.[3]

Solutions : Existing buildings must be evaluated for the presence of hazardous materials and consider their removal when technically feasible. If not possible, hazards must be isolated and periodically monitored to prevent human exposure. Generating dust that can become suspended and respirable must be avoided. Any waste generated from construction, or any other onsite activity must be handled appropriately in accordance with best practices.

PART 1 MANAGE ASBESTOS HAZARDS

For All Spaces:

Option 1: Asbestos risk assessment and remediation

The following requriements are met:

a. The building was constructed or last renovated before the enactment of laws banning the installation of asbestos-containing materials, or is located where there is no local asbestos phase-out regulation.

b. An inspector certified under local regulation or a qualified professional with demonstrable experience where no local regulations apply conducts an investigation of the project space and reports the following:

   1. A list of locations where presumed asbestos containing materials (PACM) were found.
   2. Confirmation of the presence of asbestos is performed through Polarized Light Microscopy (PLM), Scanning Electron Microscopy (SEM) or Transmission Electron Microscopy (TEM) testing. The sample number and location follow applicable laws or recommendations of the inspector conducting the assessment. Materials having over 1% of asbestos are considered ACM. If analytical confirmation is not available or possible, all PACM are considered asbestos-containing materials (ACM).

c. If asbestos-containing materials (ACM) were found per the above, an action plan that contains the following is implemented:

   1. Notification of any works to relevant authorities and persons living, working or transiting in the vicinity of the building or space.
   2. Preventative measures against the formation and spread of asbestos fibers in the air during remedial work.
   3. Measures taken for workers' protection during remediation activities, including but not limited to skin and respiratory protection.
   4. If ACM are being removed, activities are carried out for proper handling of ACM waste, including: wetting of all removed ACM, care in transportation to prevent crumbling, sealing and leak-tight transportation, proper labeling and final disposal in locations allowed by applicable laws and permits.
   5. Post-remediation clearance for occupancy confirmation by testing of fibers in air using phase contrast microscopy (PCM) or transmission electron microscopy (TEM) following standards referenced in applicable local laws or, if not available, NIOSH Manual of Analytical Methods (MNAM) Methods 7400 or 7402, GBZ/T192.5-2007, ISO 8672:2014, ISO 10312:2019 or ISO 13794:2019. The number of samples and sampling conditions must meet local regulations and/or conform to ISO 16000-7.
   6. If any of the asbestos is managed by methods other than removal, the month and year of follow-up inspection to evaluate the structural integrity of the ACM must be stated and cannot exceed three years from the date of the last inspection.

OR

Option 2: New spaces

The following requirement is met:

a. The building was constructed after the enactment of an asbestos ban in construction products.

OR

Option 3: Demonstration of prior remediation

The following requirements are met:

a. The project was last renovated after the enactment of local laws banning the installation of asbestos-containing materials.

b. The project demonstrates that asbestos remediation and clearance is a legal requirement to grant occupancy of the space.

PART 2 MANAGE LEAD PAINT HAZARDS

For All Spaces:

Option 1: Identify lead paint hazards

The following requirements are met:

a. A certified inspector or a qualified professional where no local regulations apply conducts an investigation of the space and reports the following:
   1. An inventory of locations of potential sources and sinks of lead-containing materials, where lead-containing paint may be present.
   2. Confirmation of lead hazards through in-situ test results by x-ray fluorescence (XRF) or by laboratory analyses of paint chips[4] and/or surface dusts. Surface dust is considered a hazard if its lead loading is more than 0.11 mg/m^2 of the collection area if sampled from floors or over 1.08 mg/m^2 for dust on interior window sills.[5] Paints having over 0.5% of lead by weight or 10,000 mg/m^2 of applied area and bare soil containing over 400 ppm of lead by weight are also considered lead hazards.[5] Lower thresholds mandated by local regulations prevail for terms of hazard assessment.

b. If lead is found in the investigation, a certified inspector (or a qualified professional where no local regulations apply) implements an action plan that contains the following:
   1. Notification of remediation work to occupants and transient populations in the surrounding spaces, and restriction of access to work areas during remediation.
   2. If paints are mechanically removed, measures are taken to minimize the formation and spread of dusts during the remediation process and to ensure adequate respiratory and skin protection for workers.
   3. A re-inspection schedule that includes visual assessments and dust testing, if any lead-containing paints are left in place and are subject to stabilization (i.e., painted over with products to prevent chipping or degradation) or enclosure, at least once every three years.
   4. Post-remediation clearance, confirming that the lead loading in dust is below the levels deemed hazardous.

OR

Option 2: New spaces

The following requirement is met:

a. The building was constructed after the enactment of lead paint ban.

OR

Option 3: Demonstration of prior remediation

The following requirements are met:

a. The project was last renovated after the enactment of local laws banning the application of lead-containing paint.

b. The project demonstrates through legal documentation (e.g., approved certificates of occupancy, remediation reports submitted to relevant authorities) that lead remediation was performance and clearance was provided.

PART 3 MANAGE POLYCHLORINATED BIPHENYL (PCB) HAZARDS

For All Spaces:

Option 1: PCB remediation

The following requirements are met:

a. The building was constructed or last renovated before the institution of any applicable laws banning or restricting PCBs, and is undergoing renovation work that disturbs (i.e., partially or fully removes) materials likely to contain PCBs such as caulking, fluorescent light ballasts and capacitors of appliances fabricated before 1980.

b. An inspection strategy for assessing PCB-related risks is implemented and contains the following:
   1. Determination of locations where materials potentially containing PCBs may be disturbed.
   2. If caulk is to be disturbed or removed, analysis of the presumably PCB-containing material following protocols mandated by local laws or, in absence of local laws, by any applicable US EPA[6] or ISO testing methods.

c. If PCBs are found in disturbed materials, an action plan is implemented and contains the following:
   1. Notification of remedial work to relevant authorities and building occupants.
   2. Preventative measures against the spread of PCB-containing dusts and human exposure during remediation activities, including restricting access for those not involved in the work.

3. Protective measures for workers, including chemical-resistant gloves, clothing protection, goggles and respirators.
4. Waste handling that minimizes the spread of contaminated debris and safe disposal of PCB-containing waste in locations allowed by applicable local regulations.

OR

Option 2: No PCB remediation

One of the following is met:

a. The building was constructed or last renovated before the institution of any applicable laws banning or restricting PCBs, and the project is not undergoing renovation work that disturbs (i.e., partially or fully removes) materials likely to contain PCBs such as caulking, fluorescent light ballasts and capacitors of appliances fabricated before 1980.

b. The building was constructed or last renovated after the institution of any applicable laws banning or restricting PCBs.

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X03 CCA AND LEAD MANAGEMENT | P

Intent : Mitigate risks of human exposure to chromate copper arsenate (CCA) and lead.

Summary : This WELL feature requires addressing risks associated with human exposure to chromate copper arsenate (CCA) in existing wood structures and lead in soil, playground equipment and artificial turf.

Issue : In the 2000s, the application of CCA (or 'pressure-treated wood') was restricted due to health concerns connecting its presence to arsenic exposure to humans, animals and food crops, and[1] exposure to arsenic is known to cause increased risk of skin, liver, bladder and lung cancers. [2] However, prior to its restriction, CCA was a widely used wood treatment for decades. As a result, CCA presence in our environment still poses as a threat to our health. Additionally, chromates which are catalogued as carcinogenic to humans[3] can be inhaled if CCA-containing wood is ever burnt. Among other existing external hazards is lead, which may be present in paints used on outdoor structures and playgrounds.[4] Lead may also contaminate soils upon chipping, in fibers of artificial turf[5] and in some loose rubber.[6]

Solutions : Identifying and remediating hazards associated with CCA and lead should reduce risk of exposure to and dispersion of contaminants in the environment through leachates. Although not all routes of exposure and bioavailability of lead are fully understood,[6] ingestion and inhalation of lead-containing particles may occur due to contaminated paint or rubber crumbs and, thus, from a precautionary standpoint, testing for lead is recommended.[7]

PART 1 MANAGE EXTERIOR CCA HAZARDS

For All Spaces:

Option 1: CCA assessment and remediation

For all existing wood structures installed before the enactment of laws banning chromated copper arsenate (CCA) which lie outside the building envelope but within the project boundary where human presence is expected (e.g., wooden decks, fences near walkways, playgrounds and outdoor furniture), the following requirements are met:

a. Identify CCA-containing wood through one of the following:
   1. Inspection of purchase records.
   2. Determination of whether legal bans for CCA apply.
   3. Testing for the presence of arsenic in the wood or the soil bearing the wooden structures.

b. Address CCA-containing woods through one of the following:
   1. Dispose of CCA-containing woods following applicable laws, without incinerating nor wood chipping.
   2. Treatment with penetrating (non-film-forming), oil-based, semi-transparent stains that prevent arsenic leaching on a regular basis as recommended by the manufacturer.

OR

Option 2: CCA assessment not required

One of the following is met:

a. All existing wood structures that lie outside the building envelope but within the project boundary where human presence is expected (e.g., wooden decks, fences near walkways, playgrounds and outdoor furniture) were installed after the enactment of laws banning chromated copper arsenate (CCA).

b. The project does not have wood structures that lie outside the building envelope but within the project boundary.

c. The project does not have spaces outside the building envelope but within the project boundary.

PART 2 MANAGE LEAD HAZARDS

For All Spaces:

Option 1: Lead assessment

The project addresses lead hazards through the following:

a. The top 1.5 cm layer in all existing outdoor bare soil (outside the building envelope, post-construction, not covered by grass, vegetation or other landscaping including mulch covered soil) is tested for lead. Each continuous area of bare soil is sampled at least once. If the lead concentration of any sample surpasses 400 ppm by weight,[8] then the following is performed:
   1. A second set of samples is taken at 15 cm, 30 cm, 45 cm and 60 cm deep.[9]
   2. If these samples are above 400 ppm by weight, soil is replaced with soil from another source to the extent of the deepest sample found above this threshold.

b. Lead in existing artificial turf fibers is assessed as follows:[5]
   1. If lead concentration of synthetic turf fibers is unknown, test a sample of fibers to determine the lead concentration using an EPA, ISO or locally accepted protocol.
   2. If the total lead concentration of synthetic turf fibers is greater than 300 mg/kg, perform dust-wipe testing per EPA, ISO or locally accepted protocol for dust-wipe testing to determine the surface dust-lead loading.
   3. If the wipe-testing results show total lead loadings greater than 430 μg/m^2, replace with turf containing lead concentrations less than 300 mg/kg.

c. If existing loose-fill rubber from recycled tires is present on playgrounds, sporting fields, or other surfaces, the surface is assessed and remediated per the following:

1. Sample the loose-fill rubber using an EPA, ISO or locally accepted protocol for lead testing and perform lead content analysis.
2. If the loose rubber results show total lead loadings greater than 300 mg/kg of rubber, replace the loose-fill rubber.

d. Paint applied to existing playground equipment, installed and painted before the enactment of banning laws, is assessed for lead and removed, as necessary, per the guidance below:

1. Assess the integrity and age of the paint. If the paint is cracked, peeled or chipped collect a sample for laboratory analysis for lead. Follow guidelines and methods described by the World Health Organization[10] or local equivalents for sampling and laboratory analysis.
2. Remove or encapsulate the paint from the playground equipment if the sample contains lead at a concentration over 90 ppm. Removal duties must be performed by a certified specialist or someone with demonstrable experience where no local regulations apply.

e. Newly installed bare soil, artificial turf fibers and rubber fills are certified by the manufacturer or by a third party to not contain lead above the remediation thresholds stated above.

OR

Option 2: Lead assessment not applicable

The following requirements are met:

a. The project does not have existing post-construction outdoor bare soil (e.g., not covered by grass, vegetation or mulch).

b. The project does not have artificial turf.

c. The project does not have loose-fill rubber from recycled tires.

d. Paint applied to existing playground equipment was installed and painted after the enactment of banning laws, or no playground equipment is present.

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X04 SITE REMEDIATION | O (MAX : 1 PT)

Intent : Promote the safer development of potentially contaminated sites by assessing and mitigating hazards.

Summary : This WELL feature requires site assessment, testing and remediation for the development of contaminated sites.

Issue : Contaminated soil, usually associated with past industrial activities, can leach toxic chemicals into nearby groundwater or surface waters, accumulate in sediments, volatilize and pose hazards to indoor air in buildings on the premises or be carried by wind-borne dust.[1] When left unmanaged, contaminants from such sites can pose risks to those who live and work nearby through inhalation, ingestion or dermal contact.[2] Contamination in these sites, which are known as brownfields, [1] may complicate development if not properly addressed.

Solutions : Site assessment and remediation can reduce risk of exposure to populations that live in proximity to contaminated sites. Cleanup of contaminated sites that can present environmental (e.g. air, water, soil) and human health hazards helps protect the public from associated hazards and encourages environmentally responsible growth,[3] further preserving undeveloped land.

PART 1 ASSESS AND MITIGATE SITE HAZARDS (MAX : 1 PT)

For All Spaces:

The following requirements are met:

a. Site was used for or affected by past or present industrial activities (e.g., hazardous waste storage, fuel station, manufacturing plant, on-site dry cleaners, automotive repair or brownfields).

b. Site is assessed for potential contamination in soil or underground water from past uses or surrounding conditions using one of the following:

   1. Local applicable regulation for environmental site assessments.
   2. Guidelines provided in the standard ASTM E1527-05 (Phase I site assessments).

c. If the site investigation establishes the potential presence of contaminants, the project implements a sampling strategy to quantify contamination and determine remediation needs according to local regulations or ASTM E1903-97 (Phase II site assessment) guidelines.

d. If remediation is required, the project implements a sustainable remediation plan before, during and after construction that integrates the following:[3,4]

   1. A risk-based approach to sustainable remediation (risk assessment/risk-benefit analysis).
   2. A tiered approach to assessment and an appraisal of remediation options.
   3. Safe working practices for workers during remediation.
   4. Record keeping of decision-making and assessment processes.
   5. Protocol for engaging stakeholders, including management of the impacts on the community.

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X05 ENHANCED MATERIAL RESTRICTIONS | O (MAX : 2 PT)

Intent : Minimize the exposure to certain chemicals by limiting their presence in products.

Summary : This WELL feature requires restricting chemicals found in products commonly installed in buildings.

Issue : The materials industry has been able to develop and adapt products to satisfy the needs of the market. However, some newly introduced chemicals that pose to be advantageous from a performance or cost perspective, may be associated with negative health effects. Populations that prove to be the most vulnerable to these chemicals are unborn and young children, as well as pregnant women.[1,2] Chemical classes with compounds suspected or proven to pose health concerns include orthophthalates (common plasticizers), halogenated flame retardants (HFR)[1,3] and other per-fluorinated compounds (PFCs) and heavy metals, in addition to formaldehyde.[4] Common pathways of exposure to these compounds are through inhalation, skin contact or swallowing of dusts, soils or larger particles. Overall, there is a great research gap in assessing the safety of these and many other chemicals,[5] driven both by their ubiquity in the environment, as well as their presence in the human body. For instance, Per- and polyfluoroalkyl substances (PFAS) such as perfluorooctanoic acid (PFOA) and perfluorooctanesulfonic acid (PFOS) are persisting PFCs in natural environments (including sources of drinking water) and found in humans.[6] Metabolites of orthophthalates are partially metabolized by the body and commonly found in urine.[7]

Solutions : Selecting products that are devoid of or have reduced amount of chemicals associated with health concerns may help to prevent exposure to these compounds. Some organizations have published regulations, guidelines or certifications that restrict or minimize the presence of certain chemicals from these classes.

PART 1 SELECT COMPLIANT INTERIOR FURNISHINGS (MAX : 1 PT)

For All Spaces:

1: Furniture, millwork and fixtures

At least 50% by cost of newly installed furniture, millwork and fixtures (minimum 10 distinct products), as defined in Appendix X1, meet one of the following requirements:

a. Textiles (i.e., fabrics including upholstery) and plastics in products contain 100 ppm (0.01%) by weight or less of the below compounds and chemical classes, unless higher amounts are mandated by local codes. For assessing compliance of a product, all pieces of each of the two material categories (textiles, plastics) are grouped together and each material category is assessed independently against the 100 ppm threshold:
   1. Halogenated flame retardants (HFR).
   2. Per- and polyfluoroalkyl substances (PFAS).
   3. Lead.
   4. Cadmium.
   5. Mercury.

b. Do not contain textiles and plastic.

2: Electrical and electronic products

All newly installed electrical and electronic products, as specified in Appendix X1, meet the following requirement:

a. Products are compliant with RoHS restrictions.

PART 2 SELECT COMPLIANT ARCHITECTURAL AND INTERIOR PRODUCTS (MAX : 1 PT)

For All Spaces:

At least 50% by cost of newly installed products under the classes listed below, as defined by Appendix X1 (minimum 10 distinct products), meet the following requirements, unless higher amounts are mandated by local code:

a. Flooring products contain 100 ppm (0.01%) by weight or less of the following:
   1. Halogenated flame retardants (HFR).
   2. Per- and polyfluoroalkyl substances (PFAS).
   3. Orthophthalates.

b. Insulation products, including thermal and acoustic insulation in walls, ceilings, ducts, tubes and pipes, contain 100 ppm (0.01%) by weight or less of halogenated flame retardants (HFR).

c. Ceiling and wall finishes and demountable wall partitions contain 100 ppm (0.01%) by weight or less of the following:
   1. Halogenated flame retardants (HFR).
   2. Orthophthalates.

d. Pipes and fittings intended for drinking water distribution and delivery contains 100 ppm (0.01%) by weight or less of orthophthalates.

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X06 VOC RESTRICTIONS | O (MAX : 4 PT)

Intent : Minimize the impact of volatile organic compounds (VOCs) emitted by products on indoor air quality.

Summary : This WELL feature requires adherence to emission thresholds for materials placed inside the building envelope.

Issue : VOCs encompass a wide group of volatile substances of both natural and artificial origins which have a wide range of health effects from nose, eye and throat irritation, headaches and nausea to liver, kidney and central nervous system damage.[1] Furthermore, some select VOCs are known or suspected carcinogens.[1] While these compounds are present outside, buildings are a net source due to human activities, cleaning practices and emissions from materials.[2] Newly installed furniture, insulation, flooring as well as wet-applied products such as paints, adhesives, sealants and coatings can significantly introduce VOCs into living spaces[3] for approximately one to two years,[4] and sometimes causing elevated concentrations in enclosed spaces like wardrobes.[5] These compounds can react in many ways to form new compounds and respirable particles, even seemingly impacting air pollution outdoors.[2]

Solutions : The selection of products with low or no VOC emissions is instrumental to prevent worsening in air quality. When VOCs are emitted, careful material selection in newly built spaces has been shown to accelerate VOCs reduction (i.e. off-gassing) to background levels.[6] Reducing level of toxic compound emission will also help to reduce the demands of ventilation.[4] Off-gassed products (e.g., reused furniture) also limit further emissions.

PART 1 LIMIT VOCS FROM WET-APPLIED PRODUCTS (MAX : 2 PT)

For All Spaces:

Newly installed interior wet-applied paints, coatings, adhesives, and sealants (minimum 10 distinct products or applied to at least 10% of project area) meet the following:

a. All products are tested to meet methods and thresholds established in one of the following standards and/or regulations for VOC content:
   1. SCAQMD Rule 1168 (Adhesives and Sealants, 2017).
   2. GB 33372-2020 (Adhesives).
   3. 2019 CARB SCM for Architectural Coatings.[7]
   4. EU Ecolabel for indoor and outdoor paints and varnishes.
   5. HJ 2537-2014 (Paints).
   6. Any other compliance path listed in the 'VOC content evaluation' section of the 'Low-Emitting Materials' credit of the LEED v4.1 standard.[8]

b. At least 75% of products (by surface area or volume) are tested by a third-party laboratory to meet testing methods and thresholds established in one of the following standards and/or regulations for VOC emissions:
   1. California Department of Public Health (CDPH) Standard Method v1.2.
   2. AgBB.[9]
   3. European Union LCI VOC thresholds[10] following EN 16516-1:2017 testing methods.
   4. Any compliance path accepted to meet the VOC emission requirements of the 'Low-Emitting Materials' credit of the LEED v4.1 standard.[8]

PART 2 RESTRICT VOC EMISSIONS FROM FURNITURE, ARCHITECTURAL AND INTERIOR PRODUCTS (MAX : 2 PT)

For All Spaces:

Products within one or more categories and corresponding thresholds in Table 1 meet one of the following compliance requirements, earning points as shown in Table 2: Table 1:

Product Category (from Appendix X1)	Threshold for Compliance
Flooring	90% of cost or surface area
Furniture, millwork and fixtures	75% by cost
Insulation, ceiling and wall panels	75% by cost or surface area

Table 2:

Tier	Achievement	Points
1	One compliant product category	1
2	At least two compliant product categories	2

a. Tested per methods and VOC emission thresholds established in one of the following:
   1. California Department of Public Health (CDPH) Standard Method v1.2.
   2. AgBB.[9]
   3. European Union LCI VOC thresholds[10] following EN 16516-1:2017 testing methods.
   4. ANSI/BIFMA e3-2014, sections 7.6.1 or 7.6.2 (Furniture).
   5. Any compliance path accepted to meet the VOC emission requirements of the 'Low-Emitting Materials' credit

of the LEED v4.1 standard.[8]

b. Made exclusively with one or a combination of (without organic additives): metal (including powder-coated, plated and anodized materials), untreated wood and plant products, glass, ceramic, concrete or stone.

c. If custom-made or refurbished, wet-applied and wood-based materials used in fabrication or refurbishing meet the following:

 1. All paints, coatings, sealants and adhesives applied to the product are verified as low-VOC emitting by one of the applicable standards listed in Part 1.
 2. All composite wood panels, including medium-density fiberboard, plywood and particle wood panels meet the 'Formaldehyde emissions evaluation' criterium of the 'Low-Emitting Materials' credit of the LEED v4.1 standard,[8] or meet one of the following: US EPA TSCA Title VI, Europe E1, Japan Four-star.

d. Installed for at least 6 months before enrollment or the start of subscription or manufactured and unmodified at least one year before enrollment or the start of subscription.

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X07 MATERIALS TRANSPARENCY | O (MAX : 3 PT)

Intent : Promote material transparency across building material and product supply chain.

Summary : This WELL feature requires the compilation and availability of product descriptions, with ingredients evaluated and disclosed through transparency programs.

Issue : The global supply chain for material production is multi-tiered and complex, and technical and chemical knowledge throughout the supply chain varies greatly. As a result, there is a lack of robust data and knowledge about different chemicals and their effects on human health.[1] Such limitations in awareness prevents the adoption and use of chemicals believed to be safer in the industry.[2] Additionally, building and construction materials are not required to have complete ingredient lists, which makes it difficult to make fully informed choices when selecting safer products.

Solutions : Growing scientific and public concern over chemical exposure has prompted the introduction of a number of disclosure tools to help differentiate safer ingredients and products. Labels that promote material ingredient disclosure encourage supply chain transparency and work to bridge the information gap between manufacturers and users. Further, promoting awareness of and education on material ingredient content through product labeling can help enable informed decision making.

PART 1 SELECT PRODUCTS WITH DISCLOSED INGREDIENTS (MAX : 1 PT)

For All Spaces:

For at least 25 distinct, permanently installed products (including flooring, insulation, wet-applied products, lighting fixtures, ceilings, and wall assemblies and systems), furniture and task and floor lamps, ingredients are disclosed by the manufacturer, a disclosure organization or a third party through one of the following:

a. A Declare label, operated by the International Living Future Institute.[3]

b. A Health Product Declaration (HPD) published in the HPD Public Repository, operated by the Health Product Declaration Collaborative.[4]

c. A Cradle-to-Cradle Certified™ product, a C2C Certified® Circularity product, or a product with a Material Health Certificate from the Cradle to Cradle Products Innovation Institute.[5]

d. A Product Lens Certification™, operated by UL.[6]

e. A Product Health Declaration, operated by Global Green Tag.[7]

f. A BIFMA Level scorecard compliant with BIFMA e3-2019 standard that demonstrates achievement of 4 points or more in credits 7.5.1.1, 7.5.2.2, or achievement of Option 1 in credit 7.5.3.

g. A building product declaration listed in the eBDV database, for a product classified as 'Recommended' by the Byggvarubedömningen criteria, version 7.1 or more recent.

h. A manufacturer's inventory containing CAS numbers of all individual compounds down to 1,000 ppm (0.1%). If the product contains a trade secret compound, GHS hazards of category 1 or 2 are listed and a concentration range is provided for each undisclosed component.

PART 2 SELECT PRODUCTS WITH ENHANCED INGREDIENT DISCLOSURE (MAX : 1 PT)

For All Spaces:

For at least 15 distinct permanently installed products (including flooring, insulation, wet-applied products, lighting fixtures, ceilings, and wall assemblies and systems), furniture and task and floor lamps, the following requirements are met:

a. All ingredients are disclosed down to 100 ppm.

b. All ingredients are publicly disclosed by the manufacturer, a disclosure organization or a third party through one of the following:

   1. A Declare label, operated by the International Living Future Institute.[3]
   2. A Health Product Declaration (HPD) published on the HPD repository.[4]
   3. A building product declaration listed in the eBDV database, for a product classified as 'Recommended' by the Byggvarubedömningen criteria, version 7.1 or more recent.
   4. Manufacturer's disclosure and/or through a third-party materials database platform. If the product contains a trade secret compound, GHS hazards of category 1 or 2 are listed and a concentration range is provided for each undisclosed component.

PART 3 SELECT PRODUCTS WITH THIRD-PARTY VERIFIED INGREDIENTS (MAX : 1 PT)

For All Spaces:

For at least 15 distinct permanently installed products (including flooring, insulation, wet-applied products, lighting fixtures, ceilings and wall assemblies and systems), furniture and task and floor lamps, the following requirements are met:

a. All ingredients are disclosed through one of the following:

   1. A Declare label, operated by the International Living Future Institute.[3]

2. A Health Product Declaration (HPD) published in the HPD Public Repository, operated by the Health Product Declaration Collaborative.[4]
3. A Cradle-to-Cradle Certified™ product, a C2C Certified® Circularity product, or a product with a Material Health Certificate from the Cradle to Cradle Products Innovation Institute.[5]
4. A Product Lens Certification™, operated by UL.[6]
5. A Product Health Declaration, operated by Global Green Tag.[7]
6. A BIFMA Level scorecard compliant with BIFMA e3-2019 standard that demonstrates achievement of 4 points or more in credit 7.5.2.2, or achievement of Option 1 in credit 7.5.3.

b. Ingredient disclosure is verified by a third party (i.e., an organization other than the manufacturer that is not affiliated with the ingredient disclosure certificate).

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X08 MATERIALS OPTIMIZATION | O (MAX : 2 PT)

Intent : Promote the selection of products that have been audited to minimize impacts on human and environmental health.

Summary : This WELL feature requires screening and labeling of products in accordance with programs that audit and restrict the use of hazardous ingredient contents in materials and products.

Issue : The vast quantity and variety of chemicals flowing through the global economy makes the task of tracing possible environmental and human health impact extremely difficult. In response to growing concerns over hazardous material and product ingredients, a number of screening and certification schemes have been introduced to the market to help differentiate safer alternatives.[1]

Solutions : Screening and certification schemes that analyze and restrict the use of hazardous ingredients in building materials— those that are environmental contaminants and/or pose human health hazards—can mitigate exposure to potentially harmful substances, help increase the demand for these products and ultimately catalyze market transformation.[2]

PART 1 SELECT MATERIALS WITH ENHANCED CHEMICAL RESTRICTIONS (MAX : 1 PT)

For All Spaces:

Option 1: Materials selection

For at least 25 distinct permanently installed products (including flooring, insulation, wet-applied products, ceiling and wall assemblies and systems), furniture and task and floor lamps, the following requirements are met:

a. Have ingredients inventoried to 100 ppm.

b. Meet one of the following:

1. Product is free of compounds listed in the Living Building Challenge's Red List v.4.0.[3]
2. Product meets the chemical thresholds in the Cradle to Cradle Basic Level Restricted Substances List, version 4.[4]
3. Product does not contain compounds listed in REACH Restriction, Authorization and SVHC lists.
4. Product meets an optimization path listed under 'Advanced Inventory & Assessment' in Option 2 of LEED v4.1 credit 'Building Product Disclosure and Optimization - Material Ingredients'.[5]

OR

Option 2: Future purchase of compliant products

For projects with less than 25 distinct newly and permanently installed products (including flooring, insulation, wet-applied products, ceiling and wall assemblies and systems), furniture and task and floor lamps, the following requirement is met:

a. Products purchased for future repair, renovation or replacement of building materials comply with chemical restrictions of Option 1 'Materials Selection'.

Note : For recertification, projects must provide product specification sheets for purchases of eligible products occurring after initial certification.

PART 2 SELECT OPTIMIZED PRODUCTS (MAX : 1 PT)

For All Spaces:

At least 15 distinct permanently installed products (including flooring, insulation, wet-applied products, ceiling and wall assemblies and systems), furniture and task and floor lamps, are certified under one of the following programs:

a. Cradle to Cradle Certified™ products with a Silver, Gold or Platinum level in the Material Health category or products with a Silver, Gold or Platinum level Material Health Certificate from the Cradle to Cradle Products Innovation Institute.[6]

b. Living Product Challenge, Materials and Health & Happiness Petals or Living Product Certification, operated by the International Living Future Institute.[7]

c. Global GreenTag Product Health Declaration, with a GreenTag HealthRATE™ mark at SilverHEALTH, GoldHEALTH or PlatinumHEALTH level, operated by Global GreenTag Intl. Pty Ltd.[8]

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X09 WASTE MANAGEMENT | O (MAX : 1 PT)

Intent : Mitigate environmental contamination and associated exposure to hazards present in certain wastes.

Summary : This WELL feature requires the safe management and minimization of wastes associated with hazardous chemicals present in commonly used products.

Issue : Some products may create hazardous waste if handled, transported or disposed of in an uncontrolled manner, increasing risks of exposure to contaminants released to the environment. For instance, unmanaged products containing mercury and other heavy metals may expose people to elevated toxic metals through soil, air and water.[1,2] Leftover pesticides that have become obsolete or otherwise unusable may be disposed in general-purpose dumps, and improper disposal can result in physical injury, environmental pollution and land degradation.[3] Finally, electronic waste, if not properly handled, may create significant health effects downstream. [4]

Solutions : A protocol for handling and minimizing hazardous wastes, which involves separation of hazardous from other solid wastes and procuring adequate receptors for recycling or final disposal, can help mitigate chemical pollution and associated health concerns. By raising awareness and properly managing hazardous wastes, as well as by selecting products that are easier to reuse and have a lower impact on human health, projects may minimize the generation of such wastes and the release of hazardous materials into the environment.

PART 1 IMPLEMENT A WASTE MANAGEMENT PLAN (MAX : 1 PT)

For All Spaces:

For all batteries, pesticides, lamps that may contain mercury, other mercury-containing equipment (including thermostats and thermometers),[5] and electrical and electronic equipment[6] present or expected to be present within the project during the building operations, a waste management plan that contains the following is developed and implemented:

a. Identification of roles, responsibilities and vendors for implementing the plan.[7]
b. Identification of the sources of waste, estimation of rates of generation and strategies to minimize waste generation.[7]
c. Strategies for waste collection. Each of the categorized wastes is separately contained in clearly labeled receptacles and removed from the building within one year.[5]
d. Protocols for cleaning spills of mercury (including broken fluorescent lamp tubes), pesticides and battery electrolyte fluid, including sealed containment of residues, as applicable.[5]
e. Protocols to track, measure and report waste stream flows.[7]
f. Protocols for off-site shipment of wastes.

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X10 PEST MANAGEMENT AND PESTICIDE USE | O (MAX : 1 PT)

Intent : Reduce the presence of pests in buildings primarily through integrated pest management (IPM) principles, favoring non-toxic pest control and the use of pesticides less hazardous to humans.

Summary : This WELL feature requires using IPM for pest control to reduce the application of pesticides and, when necessary, select low-hazard pesticides accompanied by signage detailing pesticide information at the site of application.

Issue : The presence of pests in crops, gardens and buildings has deleterious effects to the environment, our food supply and our health. While pesticides are tailored to address many problematic weeds, fungi, insects and rodents, exposure to specific pesticides may present danger to human health. For instance, studies have shown increased likelihood of cancer in children[1] and breast cancer[2], as well as birth defects upon maternal exposure to certain compounds. [3]

Solutions : Biological or chemical pesticides should only be used when absolutely necessary. Pesticide use, and associated risks, can be reduced through the application of IPM[4], which involves a decision-making process for the identification of pests, an understanding of the triggers that drive infestation, and the establishment of cultural, physical and educational barriers against their ingress.[5-7] When pesticides are needed, those deemed more protective of human health are preferred along with signage detailing pesticide information at the site of application providing further safeguard.

PART 1 MANAGE PESTS (MAX : 1 PT)

For All Spaces:

Option 1: Pest management development and implementation

A management plan for pest control based on integrated pest management (IPM) principles is implemented for all indoor and outdoor spaces, addressing the following:

a. Plan contains the following elements:[5-7]
   1. List of roles and responsibilities for the program development, implementation, maintenance and education.
   2. Pest management objectives, including protocols for identification of pests and metrics of progress.
   3. Design and operational measures to prevent conditions that may attract pests (e.g., access to food, water, harborage and entrance through the building envelope).
   4. Pest tolerance thresholds and control strategies (including methods and response times) for when tolerance thresholds are exceeded, attending to the safety of the applicator, the occupants and the environment.
   5. Records of pest monitoring data, pest events, pesticide applications, control actions and emergency responses.

b. Each pesticide used for periodic (i.e., non-emergency) application is listed in the plan and meets one of the following:
   1. Evaluated through the City of San Francisco Pesticide Hazard Screening Protocol with a Hazard Tier ranking of 3 (least hazardous).[8]
   2. Listed in the most recent version of the City of San Francisco's Reduced Risk Pesticide List as directed in the list (including limitations).[9]
   3. All active substances catalogued as ‘low-risk’ in the EU Pesticides Database.[10]
   4. All active substances are marked as “Approved” in the EU Pesticides Database[10] and are either classified as Class U or not classified in the latest version of “The WHO Recommended Classification of Pesticides by Hazard and Guidelines to Classification.”[11]

c. For pesticide application (periodic or emergency) within the project, the plan includes the following provisions:[12]
   1. Paper or digital notification to all building occupants on the protocol for pesticide use.
   2. Notification to all building occupants at least 24-hours prior to pesticide application, and signage posted at the site of application at least 24-hours prior to application until at least 24 hours after application.
   3. Notifications include the pesticide name, registration number, treatment location and date of application and applicator. If emergency pesticide application is needed, information on the type of emergency or reason for unplanned use.

d. The effectiveness of the plan is evaluated on an annual basis.[12]

e. The plan, records of its implementation, Safety Data Sheets (SDSs) of pesticides and results of inspections are available to occupants and owners.

OR

Option 2: Certified pest management

The project retains a service to implement and maintain an Integrated Pest Management (IPM) program, accredited under one of the following:

a. GreenShield Certification for pest management companies.

b. GreenPro Service Certification.

c. EcoWise Certification.

d. CEPA Certification.

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X11 CLEANING PRODUCTS AND PROTOCOLS | O (MAX : 2 PT)

Intent : Provide cleaning effectiveness by selecting less hazardous products and establishing adequate cleaning protocols and practices.

Summary : This WELL feature requires the restriction of hazardous or harmful ingredients in cleaning, disinfection and sanitization products, as well as the establishment of a cleaning plan, the maintenance of a cleaning schedule and a program training for staff.

Issue : Cleaning is fundamental for keeping a healthy indoor environment. Microorganisms such as house dust mites –ubiquitously present around the world– are directly related with asthma[1] and allergy[2] development. Surfaces may host pathogens present in feces and body fluids released by sick individuals, or through contact with another contaminated surface. [3] Beyond naturally-accumulating dust, commercial cleaning products may contain ingredients that may also degrade the indoor air quality and are suspected to be hazardous to human health.[4] Some products may emit substances that irritate the nose, eyes, throat and lungs and can cause or trigger asthma attacks. [5] Moreover, the interactions between cleaning agents, microbes and public health are diverse and complex[6,7], and we are just beginning to better understand them. [6] Cleaning practices may cause additional health concerns. For instance, indiscriminate use of cleaning sprays is suspected to be a risk factor for adult asthma.[8] Similarly, lack of education on the use of gloves during wet cleaning activities may explain the high prevalence of hand dermatitis in the cleaning service industry.[9,10]

Solutions : A thorough plan for cleaning operations that considers the health of occupants and cleaning staff increases the overall efficiency of the process, while reducing environmental damage.[11] The plan must align with advise from public health agencies for disinfection requirements. [12] Along personal protective equipment (PPE), the implementation of engineering controls (e.g., ventilation) and policies is key to reduce exposure to hazards during cleaning practices.[13] The provision of cleaning products that contain less hazardous ingredients may reduce the risk of respiratory and dermal symptoms.[8]

PART 1 IMPROVE CLEANING PRACTICES (MAX : 1 PT)

For All Spaces except Dwelling Units:

Option 1: Cleaning plan development and implementation

The project develops and implements a cleaning plan that meets the following requirements:

a. Details the following:[11]

1. Extent and frequency of cleaning.
2. Cleaning responsibilities of building occupants (if any) and cleaning staff.
3. Cleaning supplies and where they can be accessed.
4. Process to evaluate and document adherence to the cleaning plan.

b. Identifies the following:

1. Surfaces that require disinfection (e.g., high-touch surfaces).
2. Frequency and/or other thresholds (e.g., number of hours, number users of a space, results from a swab test) for disinfection.
3. Applicable governmental registration and directions of use (e.g., contact time and dilution rates) for disinfectants.
4. Other non-chemical tools used for disinfection, if any.

c. States the following documentation procedures:[11]

1. Record keeping practices for cleaning and disinfection activities.
2. The chain of communications with building occupants.
3. A system to log feedback from occupants and cleaning staff.

d. Specifies the following for cleaning materials and personal protection equipment (PPE):

1. PPE requirements for general cleaning and specialized tasks (e.g., disinfection or dilution or chemicals).
2. Color-coding for reusable and disposable cleaning cloths.
3. Separate cleaning of reusable cleaning materials from other clothing or products.

e. Includes the following precautions for storage of cleaning products:

1. An identifiable, fit-for-purpose storage space in accordance with the manufacturers' directions; bleach stored away from other products.
2. Color-coding and labeling of any bleach-based and ammonia-based products, indicating they are not to be mixed with one another.

f. Specifies the following for cleaning tools and equipment:

1. HEPA rated filters for vacuum cleaners.[1]
2. If carpet and woven upholstery are present, the cleaning methodology (based on manufacturer's recommendations), favoring hot water extraction if technically feasible.
3. Protocols for cleaning, maintenance and handling of waste accumulated in equipment (e.g., used vacuum cleaner bags).

g. Includes the following operational aspects:
   1. Use of cleaning and disinfection products, including dilutions (when needed) and ventilation requirements.
   2. On-site availability of current Safety Data Sheets (SDS) of cleaning and disinfection products, in languages spoken by the cleaning staff.
   3. Precautions to avoid slip hazards during and after floor cleaning.
   4. Safe disposal of waste, including soiled cleaning materials and PPE.
h. Outlines a training program that meets the following:
   1. Training covers cross-contamination prevention via hand hygiene, PPE, cleaning cloth replacement, cloth handling techniques and carrying systems to separate clean tools from dirty ones.
   2. Training is delivered to all relevant personnel including building management, building operators and contracted cleaning staff, on an annual basis, and whenever protocols change.

OR

Option 2: Certified cleaning provider

The project or organization retains a cleaning provider certified under one of the following standards:

a. Green Seal® Standard for Commercial and Institutional Cleaning Services, GS-42, operated by Green Seal Inc.[18]
b. Cleaning Industry Management Standard (CIMS) - GB criteria, operated by ISSA.[19]
c. Nordic Swan Ecolabelling for Cleaning Services criteria, operated by Nordic Ecolabelling. [20]
d. British Institute of Cleaning Science (BISc) accredited training.

PART 2 SELECT PREFERRED CLEANING PRODUCTS (MAX : 1 PT)

For All Spaces except Dwelling Units:

The project or organization has a cleaning policy that lists all surface cleaning and disinfection products and specifies that they meet the following requirements:

a. Cleaning products as-sold meet one of the following:
   1. Are labeled as 'low-hazard' or 'safer' by an ISO Reference,[14] or by a third-party certification recognized by the local government where the building is located.
   2. Have ingredients disclosed through a Safety Data Sheet (SDS) that meets EU Regulation 2015/830 (CLP),[15] or through a disclosure document that meets California State Bill No. 258, and there are no ingredients listed in the disclosure document present at 100 ppm (0.01%) or above that are classified with the following codes and hazard statements as defined by the Globally Harmonized System (GHS): H311, H312, H317, H334, H340, H350, H360, H372.
   3. Meet Feature X08 Materials Optimization.
b. Products labeled as disinfectants meet the following:
   1. Have all antimicrobial efficacy claims registered by a governmental office and stated in their label.
   2. Utilize only active ingredients only from the following list: citric acid, hydrogen peroxide, L-lactic acid, ethanol, isopropanol, peroxyacetic acid, sodium bisulfate, chitosan.
   3. Section 2 of the SDS does not contain the following GHS codes: H311, H312, H317, H334, H340, H350, H360, H372.

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X12 CONTACT REDUCTION | O (MAX : 2 PT)

Intent : Implement strategies to reduce human contact with respiratory particles and surfaces that may carry pathogens.

Summary : This WELL feature requires projects to implement design and policy strategies to minimize some instances of contact with contaminated respiratory particles, as well as reduce the number of surfaces that are necessary to touch.

Issue : Many viral diseases, including COVID-19[1] and influenza,[2] are spread by oral or respiratory emissions of liquid particles emitted by an infected person when they cough, sneeze or even exhale.[3] Factors that may affect exposure include the size distribution of the respiratory particles, [2] humidity,[4,5] air flow[5,6] and air treatment.[5,6] While the relative influence of these factors is variable, direct exposure to particles shed by an infected individual may increase a person's odds of acquiring certain diseases.[1,7] Respiratory and fecal particles can reach surfaces either by direct deposition from the source (e.g., coughing on a surface) or indirectly through hands. On inanimate contaminated surfaces (called fomites), pathogens may survive for a number of hours or even days.[8,9] Pathogens may spread from these surfaces when touched, potentially infecting people through oral or nasal exposure. [10] Fomites have been linked to disease transmission for many common viral pathogens, including rotaviruses – the most common cause of children's diarrhea in the world – and adenoviruses, and have been associated with some strains of coronaviruses.[10,11]

Solutions : Implementing design and policy strategies aimed at reducing exposure to some particles shed by infected individuals, like establishing physical distancing among people [12,13] or providing barriers to prevent respiratory particles,[14] may slow the spread of pathogens.[15] Reimagining spaces to reduce the number of surfaces a person needs to touch, as well as implementing enhanced hygiene protocols for high-touch surfaces, may reduce the risk of pathogen transmission.[16,17]

PART 1 B REDUCE RESPIRATORY PARTICLE EXPOSURE (MAX : 1 PT)

For All Spaces except Dwelling Units:

1: Contact reduction cues

The following requirements are implemented during periods when higher incidence of respiratory disease is likely:

a. At least one of the following distancing strategies:
   1. Queuing marks to increase distance between people while waiting in line (e.g., in elevator lobbies, at check-out counters) and while using moving sidewalks and escalators, as applicable.
   2. Screens, protective furnishings or other engineering controls to reduce particle exchange at security check-ins, reception areas, check-out counters and other places with frequent interaction between occupants and a stationary worker.
   3. Self-service systems to control ingress or egress to the project (e.g., at reception desks or checkout counters).

b. At least one of the following circulation strategies:
   1. One-way hallways and corridors.
   2. Separate entry and exit doors at pedestrian building entrances.
   3. Separate entry and exit for restrooms except single-user bathrooms.

2: Contact reduction policies

The following requirements are implemented during periods when higher incidence of respiratory disease is likely:

a. All of the following in any shared spaces (e.g., meeting rooms, workspaces, communal kitchens):
   1. Strategies to increase distance among occupants.
   2. Expectations and requirements for usage of face coverings or personal protective equipment.
   3. Clearly communicated rules for occupancy to reduce respiratory particle exposure and rationale for their use.

b. The project or organization implements at least one of the following communication strategies to educate occupants about the practices implemented by the project to reduce respiratory particle exposure:
   1. Monthly communication (e.g., email, webcast) to all regular occupants.
   2. Prominent signage (physical or digital) at all building entrances and in shared spaces.

Note :

Interiors projects may count base building elevators, entries and exits towards feature requirements, even if outside of the project boundary.

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 2 ADDRESS SURFACE HAND TOUCH (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

1: Surface touch management

The following requirements are met:

a. Project offers hands-free operation (through foot, voice, sensor or personal electronic device) or implements other design strategies to avoid hand operation for at least three of the following:
   1. Regularly used pedestrian doors to the project, during regularly occupied hours.

2. Elevators.
3. All water bottle fillers, water faucets, soap and paper towel dispensers.
4. Window blinds and indoor lighting switches and/or controllers.
5. Lids of trash, recycling and reuse bins.

b. Project supports occupants in maintaining hand hygiene near the following high-touch surfaces:

1. Handrails, handlebars and other structures that support mobility and accessibility.
2. Surfaces designed to help individuals with physical and/or visual disabilities fully utilize a space (e.g., push to open door buttons, wheelchair lift controls, tactile maps or signage).

c. Project establishes and communicates rules and expectations for the usage and cleaning of shared tools and devices (e.g., photocopiers, gym equipment, communal kitchen appliances, utensils) for all regular occupants.

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X13 B FAIR LABOR IN BUILDING PRODUCTS | O (MAX : 3 PT)

Intent : Accelerate the elimination of modern slavery by selecting building products from manufacturers that advance and verify fair labor rights across their facilities and in their supply chains.

Summary : This WELL feature requires the selection of building products from manufacturers that verify that fair labor practices are implemented at their manufacturing facilities and across their supply chains.

Solutions : Around the globe, people are tricked or forced into exploitative situations that they cannot refuse or leave, suffering what is collectively known as modern slavery conditions. [1] Not counting commercial sexual exploitation, an estimated 21.2 million people (of which 13 million were children) were subject to forced labor exploitation in 2021.[2] Products used in the construction of real estate are significant contributors to this problem.[3] Due to the complexity and obscurity of supply chains, it can be nearly impossible to identify products and materials that are "slavery-free."[3]

Impact : Creating demand for "slavery-free" products, in addition to aiding in the elimination of forced labor and the worst forms of child labor, can steer the market towards making fair labor practices the baseline across the building products industry.[3] As one of the main contributors of decent work,[4] provision of a living wage is likely to improve workers' physical and mental health and has the potential of breaking the poverty cycle.[5] Going deeper into the supply chain, several certification bodies assess and certify good labor practices during the extraction and processing of common raw building materials such as cement, steel and copper. Finally, companies can assess labor conditions within their own facilities and across their own supply chains through third-party audits and/or the achievement of relevant certifications.

PART 1 SELECT PRODUCTS FROM MANUFACTURERS THAT PROVIDE LIVING WAGES (MAX : 1 PT)

For All Spaces:

For at least five manufacturers of products from the product classes listed in Appendix X1, one the following is provided:

a. Manufacturer's proof of accreditation provided by one of the following organizations:
   1. SAI (Social Accountability International) Reference, administered by Social Accountability Accreditation Services (SAAS).[6]
   2. Any certification from the Reference.[7]
   3. A living wage certification from a member of the Reference.[8]

b. Manufacturer's public statement disclosing the living wage provision that includes the following:
   1. How a living wage is provided to all people (including contractors) who work in the manufacturer's facilities.
   2. How the living wage is determined and regularly updated.
   3. How organization-wide compliance with the living wage provision is confirmed, including metrics used to measure compliance.

Note :
This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 2 SELECT PRODUCTS WITH CERTIFIED RAW MATERIALS (MAX : 1 PT)

For All Spaces:

For at least ten products from the product classes listed in Appendix X1, each made by a distinct manufacturer, one of the following is met:

a. Has one of the following:
   1. A Reference with a Silver, Gold or Platinum Social Fairness level.[9]
   2. Any Reference that includes criteria for "Minimum Entitlement including Wages," "Workplace Health and Safety," and "Modern Slavery and Human Rights including Labor Rights".[10]
   3. A Reference.[11]

b. For products containing materials listed in Appendix X2, each of those materials (up to a maximum of five materials per product) meet one of the following:
   1. Is certified under one of the applicable standards listed in Appendix X2.
   2. Is supplied by a manufacturer that meets Part 1 of this feature.

Note :
This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

PART 3 SELECT MANUFACTURERS WITH TRANSPARENCY IN SUPPLY CHAIN PRACTICES (MAX : 1 PT)

For All Spaces:

For at least five manufacturers of products from the product lasses listed in Appendix X1, one of the following is met:

a. The manufacturer is a Full Member of the Reference.

b. The manufacturer has made a public statement disclosing the following:

   1. Confirmation that at least five of the manufacturer's suppliers (or 5%, whichever is greater) have been assessed for labor conditions by third-party auditors.[13,14]
   2. Confirmation that the audits were based on semi-announced visits to the manufacturer's supplier facilities.[15]
   3. The audit criteria, sampling methodology and relevant metrics of compliance.
   4. Confirmation that the facilities were found in compliance with local regulation and the following principles:[16]
      1. Workers' freedom of movement.
      2. Workers' freedom of association and collective bargaining.
      3. Forced labor.
      4. Working hours and remuneration.
      5. Occupational health and safety.
      6. Worst forms of child labor.
      7. Sexual harassment.

Note :
This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

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APPENDIX X1:

The following denominations for product classes apply throughout the Materials concept:

1. Millwork and fixtures: Built-in cabinetry/bespoke joinery, countertops, window treatments (e.g., curtains, blinds) and window films. Does not include beddings, pillows, artwork, rugs and appliances.
2. Ceiling and wall finishes: Ceiling and wall planks and tiles, acoustical treatments, wall bases and wallcoverings including wallpaper.
3. Electrical and electronic products: Cables, electrical boxes; tubing and conduit; fire alarm notification and initiating devices (e.g., strobes, pull stations); environmental, HVAC, occupancy and motion sensors and meters; and relays, thermostats and load break switches.
4. Flooring: Carpeting, resilient flooring (e.g., sheet, tiles) and any other natural or engineered floor covering product, including finished poured flooring.
5. Furniture: Movable objects intended to support various human activities such as seating (e.g., chairs, stools, sofas), eating or working (e.g., tables, desks, workstations), and sleeping (e.g., beds). Also includes objects for holding and storage such as chests, shelves, bookcases, file cabinets and cabinetry (except custom-made or built-in), and space separations such as freestanding partition panels.
6. Interior doors and windows, including door casings.
7. Insulation: Thermal and acoustic insulation in walls and ceilings. Unless explicitly stated, this class excludes duct, tube and pipe insulation.
8. Wet-applied products: Paints, adhesives, sealants, coatings and finished poured flooring.
9. Demountable Wall Partitions: Permanently installed wall systems that are designed to be relocated without damage to the product.

APPENDIX X2:

The table below lists eligible certifications by type of material.

Material	Relevant Certifications	Scope*
Copper	The Copper Mark Certification	Upstream supply chain
	Responsible Mineral Initiative	Product and Upstream supply chain
	LME (London Metal Exchange) Responsible Sourcing	Extracting/Farming
Steel and Iron	Initiative for Responsible Mining Assurance Standard	Extracting/Farming
	LME (London Metal Exchange) Responsible Sourcing	Extracting/Farming
	Responsible Minerals Initiative	Product and Upstream supply chain
	Responsible Steel Certification	Product and Upstream supply chain
Aluminum	Aluminum Stewardship Initiative (ASI) Chain of Custody Standard	Product
	Aluminum Stewardship Initiative (ASI) Performance Standard	Upstream supply chain
Stone	Ethical Stone Register	Upstream supply chain
	Fair Stone Standard	Upstream supply chain
	XertifiX Standard-Label	Extracting/Farming and Upstream supply chain
Mica, Gypsum, and Silica	Initiative for Responsible Mining Assurance Standard	Extracting/Farming
	Responsible Mica Initiative	Extracting/Farming and Upstream supply chain
	Responsible Minerals Initiative (conformant facilities)	Product and Upstream supply chain
Concrete	Concrete Sustainability Council Certification	Product and Upstream supply chain
Timber	American Tree Farm System (ATFS)	Upstream supply chain
	Programme for the Endorsement of Forest Certification (PEFC)	Extracting/Farming and Upstream supply chain
	Sustainable Forestry Initiative (SFI)	Product, Upstream supply chain and Extracting/Farming
	Forest Stewardship Council (FSC) Chain of Custody Certification	Product, Upstream supply chain and Extracting/Farming
Textiles	Better Cotton	Product and Extracting/Farming
	Fairtrade Cotton Mark	Product and Upstream supply chain
	Fairtrade Textile Standard	Product and Upstream supply chain
	FSC Certified Viscose	Product, Upstream supply chain and Extracting/Farming
	Global Organic Textile Standard	Product and Upstream supply chain
	Good Weave	Product and Upstream supply chain
	Responsible Wool Standard	Extracting/Farming
	STeP by Oeko-Tex	Upstream supply chain
	Global Recycle Standard	Product and Upstream supply chain
Leather	STeP by Oeko-Tex	Upstream supply chain

*Certifications may be of the following scope

Product: voluntary standards established by independent, third-party groups relating to environmental, social, ethical, and product safety issues. These groups incur the cost and responsibility of certifying that products adhere to their standards and award a certification seal, or logo, that proves compliance.

Manufacturing: creation or production of goods with the help of equipment, labor, machines, tools, and chemical or biological processing or formulation.

Upstream Supply Chain: a network of individuals and companies involved in creating and delivering a product to the consumer. Links on the chain begin with the producers of the raw materials and end with the finished product to the end user.

Extracting/Farming: obtaining natural resources from the Earth's crust (or other sources), growing crops or raising livestock to use them for various purposes.